New River Middle School Living Shoreline Feasibility Study

FY 20/21

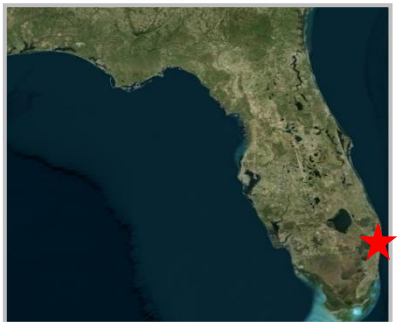
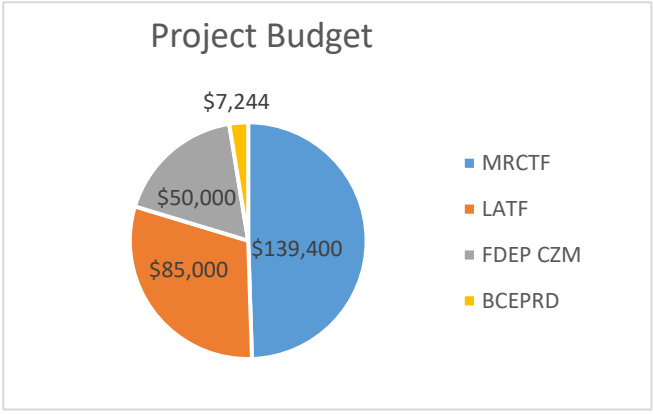


Figure 1- The New River Middle School is on the east coast in Broward County, Florida.



Project

Partners

- Florida Fish and Wildlife Conservation Commission (FWC)
- New River Middle School (NRMS)
- Broward County School Board (School Board)
- Florida Department of Environmental Protection Coastal Zone Management Program (FDEP CZM)
- Broward County Environmental Planning and Community Resilience Division (BCEPCR)

Project Description

Estuarine resources are severely depleted in Broward County because of extensive coastal development. Due to narrow waterways and excessive armoring of shorelines, few opportunities exist in the County for large-scale estuarine habitat restoration projects. The FWC, in partnership with the NRMS, and the BCEPCR, are proposing to implement a wetland restoration and living shoreline project on highly visible public lands.



Figure 2: Aerial photo of the New River Middle School

The NRMS project site has approximately 500 ft of sea-walled shoreline. This school hosts a marine science magnet program which will directly benefit from access to restored estuarine habitat on-site. The project's restoration goals are to create a living shoreline along the school's existing seawall and to enhance a portion of upland property to wetland marsh to provide research and volunteer opportunities for the NRMS students. This project is fully supported by the Broward County School District and the school's teaching and administrative staff and the surrounding community.

This project requires considerable engineering and design as it will require breaching a seawall and essentially flooding an excavated part of the NRMS's current upland property. Existing infrastructure, drainage, project stability, project performance, and school safety are all significant considerations. During fiscal years 18/19 and 19/20, $68,000 in MRCT funds were awarded to complete topography and bathymetry surveys, wetland delineation, sediment analysis, a legal boundary map, and a thorough seawall evaluation. Conceptual designs were created, and the project team met to determine the final preferred design. Following this decision, pre-application meetings were held with the regulatory agencies.

The final selected design includes three living shoreline treatments placed in front of the existing seawall. These treatments include a pre-cast concrete planter, stacked riprap, and a riprap planter. A floating dock for student access to the living shoreline is also included in the plan. The upland portion of the property will be converted to a wetland marsh via a breach in the existing seawall. The wetland will support intertidal vegetation and features a boardwalk and elevated classroom (see Figure 3).

In FY 20/21, FWC allocated $36,400 from the Marine Resources Conservation Trust Fund (MRCTF), $50,000 from the Land Acquisition Trust Fund (LATF) , and the FDEP CZM program provided $50,000 to retain WSP Consulting to advance the design and permitting needs of the project. Tasks completed include a geotechnical report, 100% design plans for the living shoreline and dock, construction cost estimates for the living shoreline and dock, 30% design plans for the wetland, and permit applications and RAI responses for all components of the project. Additionally, BCEPCR conducted benthic surveys of the site in FY 19/20 and 20/21, contributing an estimated $7,244 in in-kind services.

MRCTF and LATF funds, $35,000 each for a total of $70,000, have been awarded for FY 21/22. These funds will be used to bring the interior wetland portion of the project to 90% design and to develop construction estimates necessary to prepare grant proposals. It is anticipated that all permits will be received in FY 21/22 and the project team will begin seeking grant funds for construction of the project.

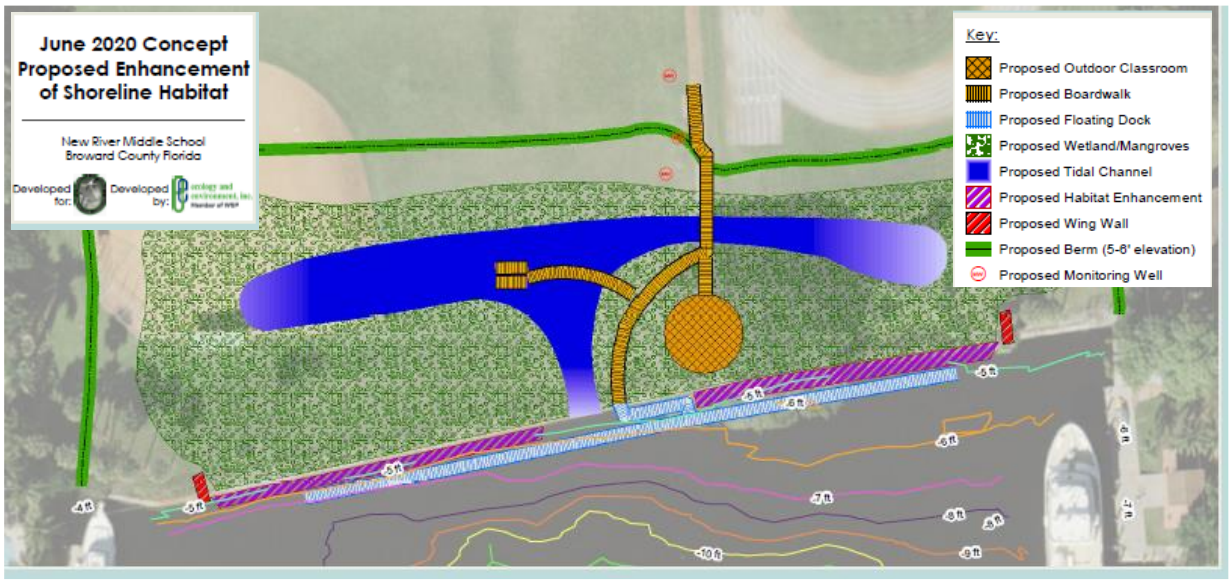


Figure 3: The preferred conceptual design selected by the New River Middle School Living Shoreline Project Team.

Benefits

Benefits to be realized from the final constructed project include:

- Creation of a mangrove living shoreline benefiting fish and wildlife and demonstrating multiple living shoreline options
- Development a living laboratory for middle school science magnet students
- Creation of a place-based living shoreline accessible to the public to demonstrate the benefits of green infrastructure and for improving coastal resiliency

MESS Strategic Plan Elements Supported

This proposal supports Objectives 1.1 and Strategic Actions 1.1.1 and 1.1.1 of Goal 1 of the MESS Strategic Plan.

Acreage restored by habitat type (footprint)

Total- 1.0 acre estuarine habitat:

- 0.5 acre mangrove and saltmarsh habitats
- 0.5 acre submerged aquatic vegetation habitat

Acreage affected by restoration activities: 87.0 acres (includes acreage listed above and 86 acres of adjacent estuarine wetland)